

ABL Customer Statement Intelligence Suite

Final Technology Stack - Completed 10 Sep 2026

Component	Selected Technology	Reason
Frontend	React.js + Tailwind CSS	Reusable UI components and rapid styling.
Build Tool	Vite	Fast React development and optimized builds.
Frontend Deployment	Vercel	Git-based production frontend hosting.
Backend	FastAPI (Python 3.12)	Typed APIs and AI/ML-friendly backend.
Backend Deployment	Heroku	Production FastAPI hosting and PostgreSQL integration.
LLM	Groq - openai/gpt-oss-20b	Final low-latency language-generation provider/model.
Embedding Runtime	FastEmbed / ONNX	Lower-memory embedding runtime for Heroku.
Embedding Model	all-MiniLM-L6-v2 (384D)	Free/local semantic embeddings.
Database	PostgreSQL	Structured and policy-corpus storage.
Vector Search	PostgreSQL + pgvector	Vector similarity search without a separate vector DB.

Final architecture: React/Vite frontend on Vercel communicates with FastAPI on Heroku. Exact financial facts are computed deterministically in Python; Groq provides grounded qualitative language generation. Policy Q&A; uses FastEmbed MiniLM embeddings with PostgreSQL + pgvector and returns public ABL sources.

Final production status: All three modules passed production smoke/regression testing on 10 Sep 2026. The current release showed no new Heroku R14 memory errors after the FastEmbed migration.

Live: <https://statement-intelligence-suite-abl.vercel.app>

Completion: 27 Aug 2026 - 10 Sep 2026 | Status: FINAL - Production Functional